

FINANCIAL DATA ANALYTICS DASHBOARD



- SQL
- VS Code
- Power Query
- Power BI

2010 – 2013 | AdventureWorksDW

Revenue & Profit Insights Report

A complete end-to-end data analytics project extracting, transforming, and visualizing bicycle sales performance across 6 countries.

80.45M

Revenue

470K

Profit

0.6%

Margin

214K

Units

From Raw Database to Executive Dashboard

This project follows a complete data analytics lifecycle across 4 structured phases:

01



SQL Extraction

Raw data from
AdventureWorksDW

02



JSON Export

Validated in
VS Code

03



Power Query

Cleaned &
Transformed

04



Power BI

Interactive
Dashboard

Dataset: 9,501 rows • 12 Columns • Period: Dec 2010 – Nov 2013 • 6 Countries • 4 Product Categories



STEP 1

Data Extraction using SQL

The screenshot shows a SQL IDE interface with a menu bar (File, Edit, Selection, View, Go, Run, ...), a search bar, and a toolbar. The main editor displays a SQL query for data extraction from a financial database. The query is as follows:

```
1 SELECT
2     r.ResellerName AS Reseller,
3     d.FullDateAlternateKey AS OrderDate,
4     YEAR(d.FullDateAlternateKey) AS SalesYear,
5     MONTH(d.FullDateAlternateKey) AS SalesMonth,
6     DATENAME(MONTH, d.FullDateAlternateKey) AS MonthName,
7     d.FiscalQuarter AS FiscalQuarter,
8     g.EnglishCountryRegionName AS Country,
9     pc.EnglishProductCategoryName AS ProductCategory,
10    SUM(f.OrderQuantity) AS Quantity,
11    SUM(f.SalesAmount) AS Revenue,
12    SUM(f.TotalProductCost) AS Cost,
13    SUM(f.SalesAmount - f.TotalProductCost) AS Profit
14 FROM dbo.FactResellerSales f
15 JOIN dbo.DimReseller r
16     ON f.ResellerKey = r.ResellerKey
17 JOIN dbo.DimDate d
18     ON f.OrderDateKey = d.DateKey
19 JOIN dbo.DimProduct p
20     ON f.ProductKey = p.ProductKey
21 JOIN dbo.DimProductSubcategory ps
22     ON p.ProductSubcategoryKey = ps.ProductSubcategoryKey
23 JOIN dbo.DimProductCategory pc
24     ON ps.ProductCategoryKey = pc.ProductCategoryKey
25 JOIN dbo.DimGeography g
26     ON r.GeographyKey = g.GeographyKey
27 GROUP BY
28     r.ResellerName,
29     d.FullDateAlternateKey,
30     d.FiscalQuarter,
31     g.EnglishCountryRegionName,
32     pc.EnglishProductCategoryName
33 ORDER BY
34     d.FullDateAlternateKey,
35     r.ResellerName,
36     g.EnglishCountryRegionName,
37     pc.EnglishProductCategoryName;
```

The status bar at the bottom indicates: Ln 1, Col 1 | Spaces: 4 | UTF-8 | CRLF | MSSQL | SQLCMD: Off | 9501 rows affected | 00:00:01.338 | prod-sql-cfeducation.database.windows.net | AdventureWorksDW | Query executed

Activate Windows
Go to Settings to activate Windows.



STEP 2

Export to JSON & Preview in VS Code

File Edit Selection View Go Run ... Search

Financial Data Analytics Dashboard Revenue, Cost & Profit Insights Financial Data Analytics Dashboard Revenue, Cost & Profit Insights, Josn.json X

C: > Users > Admin > Downloads > Financial Data Analytics Dashboard Revenue, Cost & Profit Insights, Josn.json > ...

```
1  {
2
3    "Reseller": "Area Bike Accessories",
4    "OrderDate": "2010-12-29T00:00:00",
5    "SalesYear": 2010,
6    "SalesMonth": 12,
7    "MonthName": "December",
8    "FiscalQuarter": 2,
9    "Country": "United States",
10   "ProductCategory": "Accessories",
11   "Quantity": 7,
12   "Revenue": 141.3055,
13   "Cost": 84.1946,
14   "Profit": 57.1109
15 },
16 {
17   "Reseller": "Area Bike Accessories",
18   "OrderDate": "2010-12-29T00:00:00",
19   "SalesYear": 2010,
20   "SalesMonth": 12,
21   "MonthName": "December",
22   "FiscalQuarter": 2,
23   "Country": "United States",
24   "ProductCategory": "Bikes",
25   "Quantity": 13,
26   "Revenue": 10273.9772,
27   "Cost": 10301.8837,
28   "Profit": -27.9065
29 },
30 {
31   "Reseller": "Area Bike Accessories",
32   "OrderDate": "2010-12-29T00:00:00",
33   "SalesYear": 2010,
34   "SalesMonth": 12,
35   "MonthName": "December",
36   "FiscalQuarter": 2,
37   "Country": "United States",
38   "ProductCategory": "Clothing",
39   "Quantity": 5,
```

Activate Windows
Go to Settings to activate Windows.

Ln 1, Col 1 Spaces: 2 UTF-8 CRLF Json



STEP 3

Data Cleaning & Transformation — Power Query

Data Types

- ✓ OrderDate → Date type
- ✓ SalesYear, SalesMonth → Int64
- ✓ FiscalQuarter → Int64
- ✓ Quantity → Int64
- ✓ Revenue, Cost, Profit → Decimal

Column Validation

- ✓ 205 distinct resellers
- ✓ 7 distinct order dates
- ✓ 2 distinct sales years
- ✓ 5 distinct months
- ✓ 4 distinct quarters
- ✓ 0% Errors • 0% Empty

M Code Applied

- ✓ `Table.TransformColumnTypes()`
- ✓ Promoted Headers
- ✓ Use First Row as Headers
- ✓ 999+ rows profiled
- ✓ 12 columns structured
- ✓ Loaded to Power BI

Result: Clean, structured dataset with no errors, ready for analysis and visualization in Power BI



POWER BI DASHBOARD



0.6%
Margin %

470.48K
Total Profit

80.45M
Total Revenue

79.98M
Total Cost

214K
Total Quantity

Date

12/29/2010

11/29/2013

Country

All

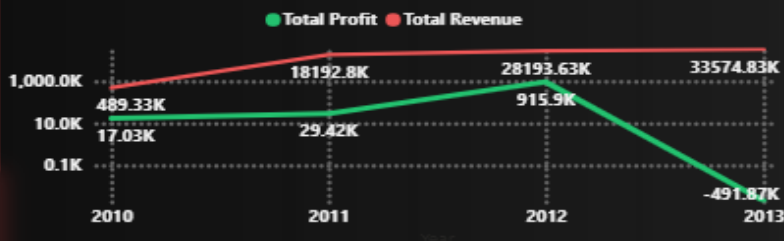
Reseller

All

Category

All

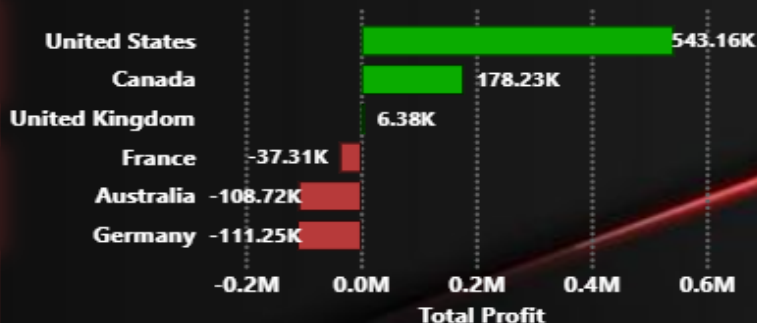
Total Profit and Total Revenue by Year



Total Quantity by Year

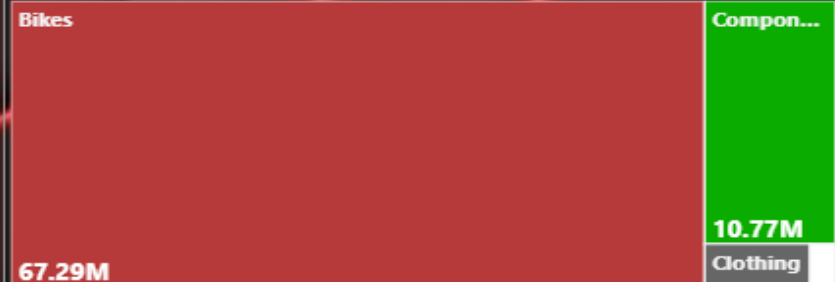


Total Profit by Country



Cost by Product Category

ProductCategory ● Bikes ● Components ● Clothing ● Accessories





Strong Revenue Growth

Revenue grew from 0.49M (2010) to 33.57M (2013), a massive ~68x increase reflecting strong market expansion over the period.



2013 Profitability Crisis

Despite record revenue of 33.57M, 2013 recorded a loss of -491.87K. Cost pressure surged, squeezing margin to near zero.



US & Canada Drive Growth

United States contributed 543K profit, Canada 178K. These two markets account for virtually all positive returns.



Germany, Australia & France Drag

Germany (-111K), Australia (-109K), and France (-37K) generated negative profits — pricing and cost strategy review needed.



Bikes Dominate Costs

Bikes category accounts for ~84% of total costs (67.29M of 79.98M), making it the single most critical lever for margin improvement.



Volume Scaling Fast

Units sold grew from 0.82K (2010) to 103.66K (2013) — a 126x volume increase, outpacing revenue growth, signaling price compression.

**THANK
YOU**